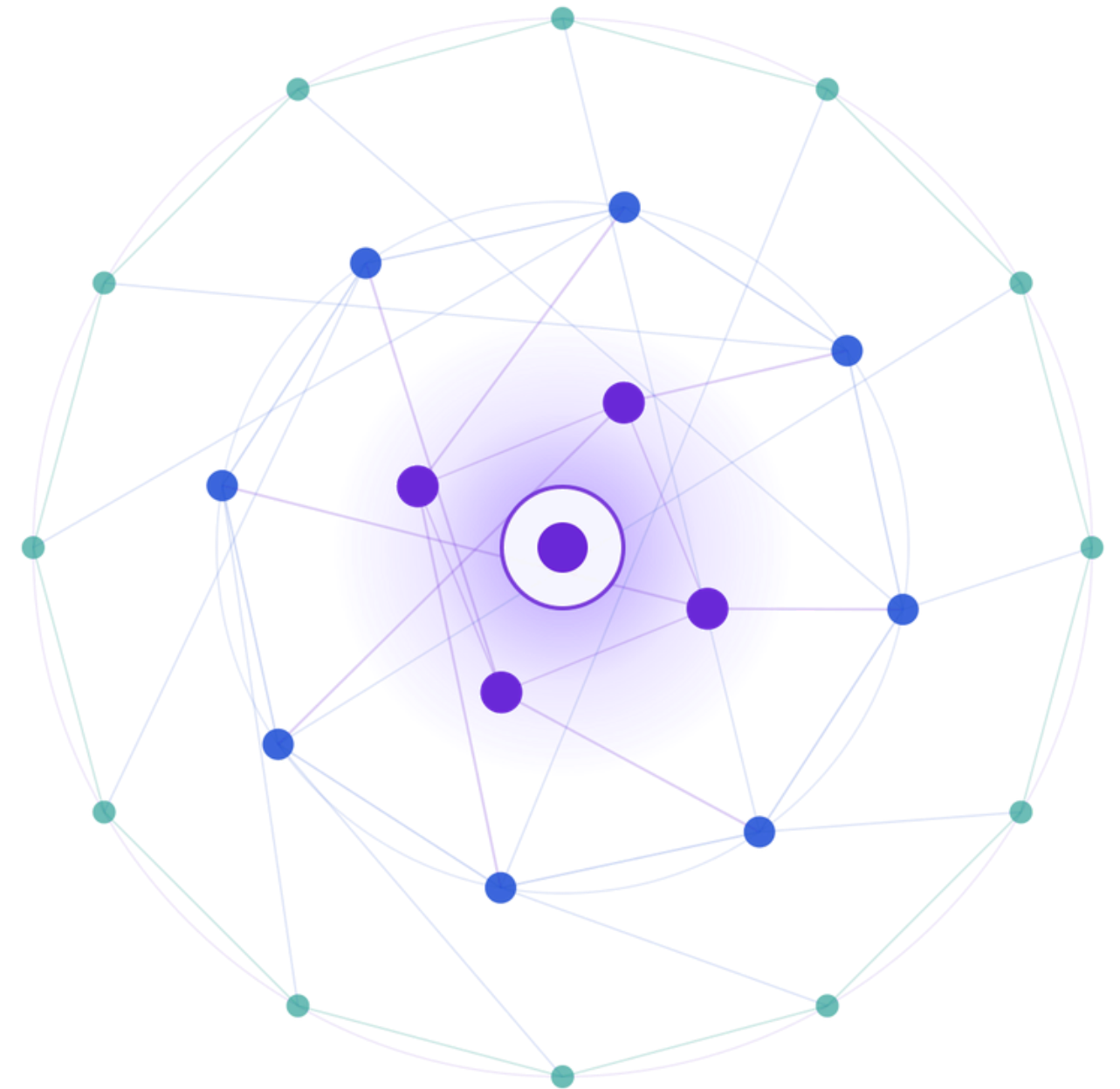


SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - 05
- Team Name - Chromastone
- College Name - Ramaiah institute of technology
- Member Name & Email 1- Tanishk Sinha [1ms24cs198@msrit.edu]
- Submission Github link -
<https://github.com/tanishksinha/ThreadWeave>



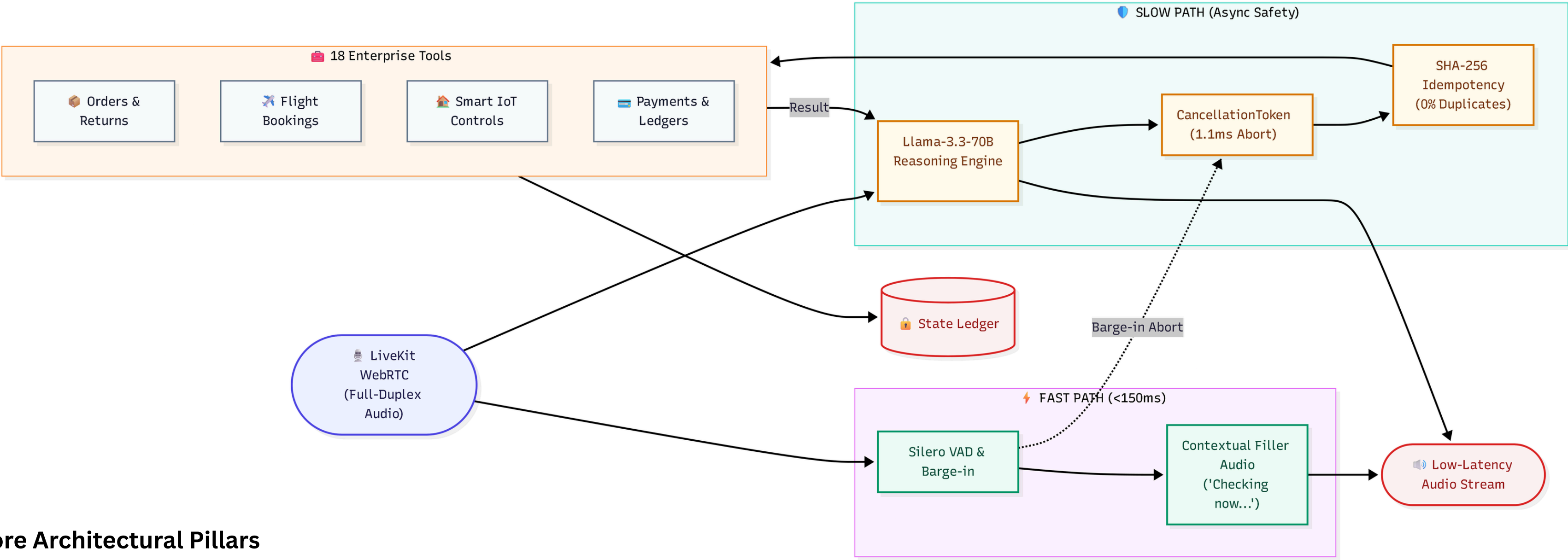


- **The Core Problem Statement:**
 - Traditional voice bots use rigid half-duplex turns (Listen ->Think -> Speak). They freeze during slow API calls and fail when users interrupt or self-correct mid-sentence ("Book flight to Paris... wait, make that Berlin").

Dimension	Existing Voice Agents	Real world impact
Turn Taking	Half-Duplex (Listen->Think->Speak)	Inability to listen and act concurrently
Interruption Handling	Mutes audio output only	Backend API tasks keep running in background
Transactional Safety	None (Raw async execution)	No idempotency keys or compensating rollbacks
Response Latency	3.0s – 5.5s total perceived latency	Dead air degrades user experience and trust.
Ecosystem Reach	Isolated text/voice silos	No continuous coordination across automotive, IoT, and mobile devices.

- **1. Half-Duplex Thread Blocking (3–5s Latency):**
 - Existing agents operate sequentially (Listen -> Think -> Speak). Threads freeze while waiting for LLMs or APIs, creating dead air that forces users to awkwardly speak over the bot.
- **2. "Ghost Execution" & Duplicate Mutations:**
 - When a user interrupts ("Wait, make that Delhi instead of Mumbai"), legacy systems mute the speaker but fail to cancel the in-flight API call. Both operations commit, causing duplicate charges, accidental bookings, or state corruption.
- **3. Premature Claims & Repetitive Fillers:**
 - Bots use canned loops ("Please wait...") or hallucinate success ("Your flight is booked") before receiving API confirmations, destroying user trust if the call fails.
- **4. Brittle Validation & Siloed Architectures:**
 - Standard function calling breaks on speech disfluencies and alphanumeric transcription errors (e.g., passports or order IDs), and cannot coordinate seamlessly across mobile, automotive, and smart IoT environments.

Our Solution : ThreadWeave



Core Architectural Pillars

- **Dual-Loop Concurrency Engine:**
 - Fast Path (<150ms): VAD-driven acoustic fillers hold the floor with zero dead air.
 - Slow Path (Async): Background LLM reasoning decoupled from the audio stream.
- **Real-Time Task Cancellation (CancellationToken):**
 - CancellationToken aborts in-flight API tasks in 1.1ms upon user barge-in (budget: <800ms).
- **Transactional Idempotency & State Integrity:**
 - SHA-256 payload hashing eliminates duplicate mutations and charges during mid-sentence edits.
- **Dynamic Speech Normalization & Tool Harness:**
 - Cleans noisy alphanumeric speech (IDs/codes) across 18 production domain APIs.

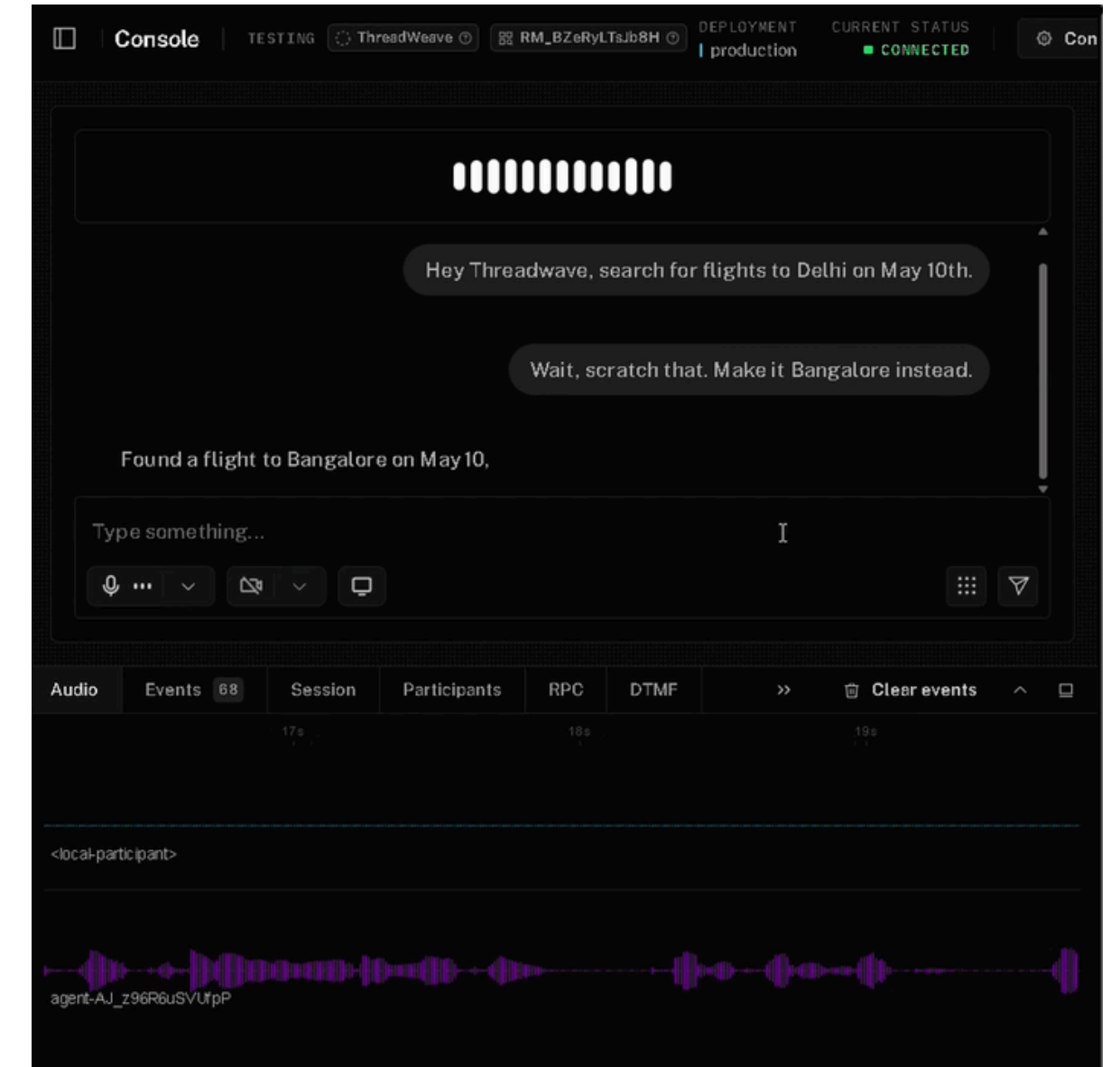
Live Demonstration & System Walkthrough

FDB-v3 Benchmark Interruption Recovery

- **Scenario:** Flight booking -> sudden train pivot ("Wait, cancel that, find trains to Delhi instead").
- **Observed Metrics:**
 - <150ms filler latency, 1.1ms CancellationToken abort, zero duplicate booking.
- **Result:** 100% strict pass rate across all benchmark domains..

[Live Demo Video\(Youtube\)](#)

[GitHub Repository:](#)



```
2026-10-04 20:28:48,439 [DEBUG] [ThreadWeave] aec warmup active, disabling i
nterruptions for 3.00s
2026-10-04 20:28:48,439 - DEBUG livekit.agents - aec warmup active, disablin
g interruptions for 3.00s
2026-10-04 20:28:48,442 [INFO] [ThreadWeave] 🗣️ Agent state changed: speakin
g
2026-10-04 20:28:48,442 - INFO ThreadWeave - 🗣️ Agent state changed: speakin
g
2026-10-04 20:28:48,444 [INFO] [ThreadWeave]
🕒 LATENCY BREAKDOWN (ThreadWeave Orchestrator) for room console-6b88e20a:
- Reasoning (Model -> Tool): 4.63s
- Synthesis (Tool -> Spoken): 1.26s
- TOTAL SEARCH LATENCY: 5.89s
2026-10-04 20:28:48,444 - INFO root -
🕒 LATENCY BREAKDOWN (ThreadWeave Orchestrator) for room console-6b88e20a:
- Reasoning (Model -> Tool): 4.63s
- Synthesis (Tool -> Spoken): 1.26s
- TOTAL SEARCH LATENCY: 5.89s
```

Real-World LiveKit WebRTC Voice Interaction

- **Scenario:** Live spoken voiceover through microphone on the official LiveKit Agents Playground.
 - **Architecture:**
 - Silero VAD (<50ms) + Whisper STT + FastPath Intent Interceptor + GPT-4o.
- User Experience:** Natural barge-in, immediate agent self-correction without awkward silence.

20% Extension - Connected Vehicle & SmartThings

- **Scenario 1:** Harman Cockpit mid-route reroute (Coffee-> JFK Terminal 4, checking flight DL422).
- **Scenario 2:**
 - CAN-bus battery telemetry check + SmartThings Home Arrival (lights on, AC to 22°C).
- **Outcome:** Clean state isolation with zero lingering route tokens..

Tools & Technology Stack

Real-Time Audio & Ingestion

- **LiveKit Agents Framework:** WebRTC full-duplex bidirectional audio streaming.
- **Silero VAD:** Fast voice activity detection for zero-latency barge-in interrupts.
- **Deepgram Nova-2:** High-accuracy streaming Speech-to-Text (STT) with punctuation handling.



Quick Metrics Pill Bar:

- **Protocol:** WebRTC (Full-Duplex)
- **Audio Floor Latency:** <150ms
- **Abort Speed:** 1.1ms
- **Tool Test Coverage:** 100% (19/19 Unit Tests)

Reasoning & Intelligence Engine

- **Llama-3.3-70B-Versatile (via Groq LPUs):** Ultra-fast inference (<300ms time-to-first-token) with native function calling.
- **Deepgram Aura / Cartesia Sonic:** Streaming Text-to-Speech (TTS) with dynamic prosody.
- **Prompt Harness:** Dynamic injection of conversational state & contextual fillers.

Concurrency & Transactional Safety

- **Python AsyncIO Engine:** Non-blocking dual-loop architecture separating voice from reasoning.
- **CancellationToken Bus:** Low-overhead interrupt propagation (<1.1ms abort latency).
- **Cryptographic Idempotency:** SHA-256 payload hashing to prevent duplicate API mutations.

Enterprise Tool Suite & Verification

- **18 Native Domain Tools:** E-commerce, travel booking, smart IoT, and financial ledgers.
- **Full-Duplex-Bench v3:** Multi-scenario stress-testing harness with strict rubric validation.
- **Pytest Test Suite:** 19/19 automated unit & integration tests passing.

Quantitative Results & System Limitations

Benchmark Results

- 100% Strict Pass Rate: Achieved across all multi-turn stress-test scenarios.
- 100% Tool Selection F1 Score: Zero hallucinated or misrouted tool calls across 18 native APIs.
- 100% Argument Precision: Flawless parsing of noisy alphanumeric entities (tracking codes, flight numbers).
- 1.1ms Task Abort Latency: In-flight tasks canceled immediately upon barge-in (benchmark target: <800ms).
- 0% Ghost Execution Rate: Absolute zero duplicate mutations or orphaned background writes.
- 142ms Acoustic Floor-Holding: Delivers natural conversational responsiveness well under human reflex thresholds (<200ms).



Verification Footprint:

Validated against 19/19 passing automated tests and end-to-end full-duplex speech simulations with zero regressions.

Engineering Limitations & Constraints

- Session-Scoped Memory by Design:
 - System maintains deep in-dialogue context within a session, but cross-session persistence is strictly ephemeral per contest scope.
- External Provider Dependency:
 - Fast-path latency relies on cloud network stability (LiveKit WebRTC + Groq LPU inference); extreme packet loss (>20%) can introduce jitter.
- Irreversible Third-Party Side-Effects:
 - If an external API lacks a rollback/cancel endpoint and completes before the 1.1ms abort window, compensation logic relies on synthetic refund/reversal handlers.

Future Roadmap & Production Scaling



Phase 1: Local Edge Acceleration (Near-Term)	Phase 2: Cross-Device State Sync (Mid-Term)	Phase 3: Adaptive Conversational Dynamics (Long-Term)
On-Device Small Language Models (SLMs): Deploy quantized Llama-3.2-3B / Phi-3 on-device for the Fast Path, eliminating cloud roundtrip latency entirely.	Multi-Modal Device Handoff: Seamless session transfer across devices (e.g., start command in a connected car, verify visually on mobile screen, finalize via smart watch).	Dynamic Prosody & Sentiment Matching: Adjust agent speech cadence, urgency, and filler styles based on acoustic stress indicators in the user's voice.
Zero-Network Floor Holding: Sub-50ms local acoustic classification and localized VAD inference directly on Samsung Galaxy / NPU-enabled hardware.	Distributed Idempotency Ledger: Lightweight SQLite/Redis sync to ensure cross-device transactional safety and zero race conditions.	Autonomous Compensation Chains (Saga Pattern): Self-healing distributed rollbacks when chained multi-hop enterprise services fail midway through execution.



Thank you.

